

Problem Solving and testing using Java

LabTask -2 Question

Course code:10211CS227

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PROBLEM STATEMENT:

Task 2 (Difficult): Real-Time Stream Analytics Engine

Problem Description

A cloud analytics platform receives millions of sensor readings. Each reading contains a sensor ID and temperature value. Using stream processing concepts, perform the following operations:

1. Filter temperatures greater than 50.
2. Group readings by sensor ID.
3. Compute average temperature per sensor.
4. Sort sensors based on average temperature in descending order.

Input Format

- First line contains integer N.
- Next N lines contain SensorID and Temperature.

Output Format

Display SensorID and average temperature sorted in descending order.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq \text{Temperature} \leq 100$

Sample Input

```
6
S1 60
S2 40
S1 80
S3 70
S2 90
S3 30
```

Sample Output

```
S1 70.0
S2 90.0
S3 70.0
```

PROGRAM:

```
import java.util.*;
import java.util.stream.*;

public class LabTask2
{
    static class Reading
    {
        String sensorId;
        double temperature;
    }

    Reading(String sensorId, double temperature) {
        this.sensorId = sensorId;
        this.temperature = temperature;
    }
}
```

```

    }
    public static void main(String[] args)
    { Scanner sc = new Scanner(System.in);
      int n = sc.nextInt();
      List<Reading> readings = IntStream.range(0, n)
        .mapToObj(i -> new Reading(sc.next(), sc.nextDouble()))
        .filter(r -> r.temperature > 50)
        .collect(Collectors.toList());
      Map<String, Double> avgMap = readings.stream()
        .collect(Collectors.groupingBy(
          r -> r.sensorId,
          Collectors.averagingDouble(r -> r.temperature)
        ));
      avgMap.entrySet().stream()
        .sorted(Map.Entry.<String,
Double>comparingByValue(Comparator.reverseOrder())
          .thenComparing(Map.Entry.comparingByKey()))
        .forEach(e -> System.out.println(e.getKey() + " " + e.getValue()));
    }
}

```

OUTPUT:

```

<terminated> LabTask2 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Aug 13
6
S1 60
S2 40
S1 80
S3 70
S2 90
S3 30
S2 90.0
S1 70.0
S3 70.0

```

